The route formulation

Lab handout 5 (Lab 2)

October 2026

Files: formulations/route_formulation_vwp.py (to be completed) and labs/lab2_route_formulation.py.

1. From links to paths

The flow formulations of handouts 3 and 4 decide how many wavelengths each demand sends on each link, and the paths of a demand appear only in the solution. The route formulation, also called path formulation, works the other way round. For every demand a small set of candidate paths is computed in advance, and the model only decides how many wavelengths of the demand follow each candidate path. In this lab the candidate paths of a demand are its k shortest paths by number of hops, computed with NetworkX as in handout 2, section 4.

The problem is the one of handout 3: all the demands are routed with the minimum number of wavelengths summed over all links, and a demand may be split over several paths. The model counts the wavelengths used on each link but not which ones, so it assumes that every node can convert a wavelength into another one (no wavelength continuity, in short VWP).

symbol	meaning	code
P_d	candidate paths of demand d: its k shortest paths by number of hops	`paths[d.id]`
l_p	number of links of path p	`len(path) - 1`
$r_{d,p}$	wavelengths of demand d on its candidate path p	`r[d.id, p]`
u_e	capacity of link e: fibres × wavelengths per fibre	`G.edges[i, j]["fibers"] * G.edges[i, j]["capacity"]`

$$
\begin{aligned}
&\textit{minimise} && \Sigma_{d \in D} \Sigma_{p \in P_d} l_p \, r_{d,p} \\
&\textit{subject to} && \Sigma_{p \in P_d} r_{d,p} = h_d, \quad \text{for all } d \in D \\
& && \Sigma_{d \in D} \Sigma_{p \in P_d:\, e \in p} r_{d,p} \le u_e, \quad \text{for all } e \in E \\
& && r_{d,p} \ge 0 \text{ and integer}
\end{aligned}
$$

There are no flow conservation constraints: every candidate path already connects the origin of its demand to the destination, so it is enough to require that the wavelengths on the candidate paths add up to the traffic of the demand. The model is therefore much smaller. For Spain-7 with the 24 demands of handout 3 it has 24 k variables and 24 + 16 = 40 constraints, against 384 variables and 184 constraints of the flow formulation. The price is that only the candidate paths can be used: when the good paths are not among them, the optimum is worse than that of the flow formulation, or the problem has no solution.

2. Completing the exercise

The file `formulations/route_formulation_vwp.py` contains the model in its first lines and five blocks to be completed, marked `TODO 1` to `TODO 5`. As in handout 4, write the code of each block and then delete its `raise NotCompletedError(...)` line. The dictionary `paths_through`, which lists for every link the pairs (demand, candidate path) that use it, is given.

TODO	what to write	hint
1	the k candidate paths of every demand, `paths[d.id]`	`nx.shortest_simple_paths` and `islice`, as in handout 2
2	a variable `r[d.id, p]` for every demand and candidate path	integer, `lowBound=0`, a unique name
3	the objective function	each variable multiplied by the number of links of its path
4	the demand constraints	they replace the flow conservation constraints
5	the capacity of every link	sum over the pairs in `paths_through[i, j]`

The formulation is tested with *Check* or with

```
python -m pynet2plan check formulations/route_formulation_vwp.py
```

As long as a block is missing, the answer is NOT_COMPLETED:

```
Test network: 5 nodes, 12 links (4 wavelengths each, 3 on D-E), 5 demands. It is designed so that a
missing capacity constraint, splitting where it is not allowed or a missing wavelength-clash
constraint are detected.

Status: NOT_COMPLETED
This formulation is not complete yet (route_formulation_vwp.py, line 77).
Write the code of this TODO, then delete the line 'raise NotCompletedError(...)'. If you already
wrote the code, just delete that line.

route_formulation_vwp.py, line 77, in solve:
    raise NotCompletedError("TODO 1: compute the k candidate paths of every demand.")
NotCompletedError: TODO 1: compute the k candidate paths of every demand.
```

When the five blocks are correct, the answer is OPTIMAL with objective 20, the value of the splittable flow formulation on the same network:

```
Test network: 5 nodes, 12 links (4 wavelengths each, 3 on D-E), 5 demands. It is designed so that a
missing capacity constraint, splitting where it is not allowed or a missing wavelength-clash
constraint are detected.

Status: OPTIMAL
Validation passed (every demand carried, flow on existing links, link capacities). This is a tiny
network: also try a real one, e.g. spain7 with a small capacity (sweep 20-30).
Objective 20.0, wavelengths used 20.0, blocked 0, time 0.05 s
```

result of *Check*	probable cause
NOT_COMPLETED	a `raise NotCompletedError(...)` line has not been deleted
ERROR: "Two variables have the same name"	the name of the variables does not contain both `d.id` and `p`
UNBOUNDED	integer variables without the lower bound `lowBound=0`
INVALID: "Demand 0 (A->C) carries 0 of the 3 requested wavelengths"	demand constraint written with `<=` instead of `==`
INVALID: "Link B->C carries 6 wavelengths but its capacity is 4"	capacity constraint missing, or not summed over `paths_through[i, j]`
OPTIMAL, with a warning on the objective	the objective does not multiply the variables by `len(path) - 1`
the script does not end on a large network	`islice` is missing, so every path of every demand is enumerated

3. Spain-7 with C = 40

The script of this lab compares the route formulation with the splittable flow formulation:

```
python labs/lab2_route_formulation.py
```

Its first part repeats the check; its second part solves Spain-7 with C = 40 on every link and the traffic matrix of handouts 3 and 4:

```
24 demands, 229 wavelengths (spain7_traffic_split_vs_unsplit.csv)
                       status objective variables constraints split
flow formulation      OPTIMAL       425       384         184     2
route, k = 1       INFEASIBLE         -        24          40     -
route, k = 2          OPTIMAL       425        48          40     2
route, k = 3          OPTIMAL       425        72          40     2
route formulation: |D| x k variables (24 x k) and |D| + |E| constraints (24 + 16); flow formulation:
|D| x |E| variables (24 x 16) and |D| x |N| + |E| constraints (24 x 7 + 16)
```

With k = 1 every demand has a single candidate path, one of its fewest-hop paths. This is the fewest-hop routing of the lower bound 408, which sends 57 wavelengths over Sevilla→Malaga (handout 4, section 4), so the problem is infeasible for every C below 57. With k = 2 the optimum is already 425, the value of the flow formulation, with 48 variables and 40 constraints instead of 384 and 184. The routes may differ from those of the flow formulation, since several routings use exactly 425 wavelengths.

4. How many candidate paths?

On Spain-7 two candidate paths per demand are enough. The third part of the script uses a network with more alternative paths, COST239 (11 nodes, 52 links, 25 demands), and solves it for C = 6 to 15:

```
COST239: 11 nodes, 52 links, 25 demands, 129 wavelengths
   C   flow   k = 1   k = 2   k = 3   k = 4
   6 infeas. infeas. infeas. infeas. infeas.
   7    213 infeas. infeas. infeas.     216
   8    209 infeas.     225     213     209
   9    207 infeas.     214     207     207
```

```
  10    205 infeas.     206     205     205
  11    203 infeas.     203     203     203
  12    201 infeas.     201     201     201
  13    199 infeas.     199     199     199
  14    197 infeas.     197     197     197
  15    196 infeas.     196     196     196
k = 1: same optimum as the flow formulation for C = none of these
k = 2: same optimum as the flow formulation for C = 11-15
k = 3: same optimum as the flow formulation for C = 9-15
k = 4: same optimum as the flow formulation for C = 8-15
figure: results/lab2_route_formulation/lab2r_objective_vs_k.png
figure: results/lab2_route_formulation/lab2r_objective_vs_k.pdf
```

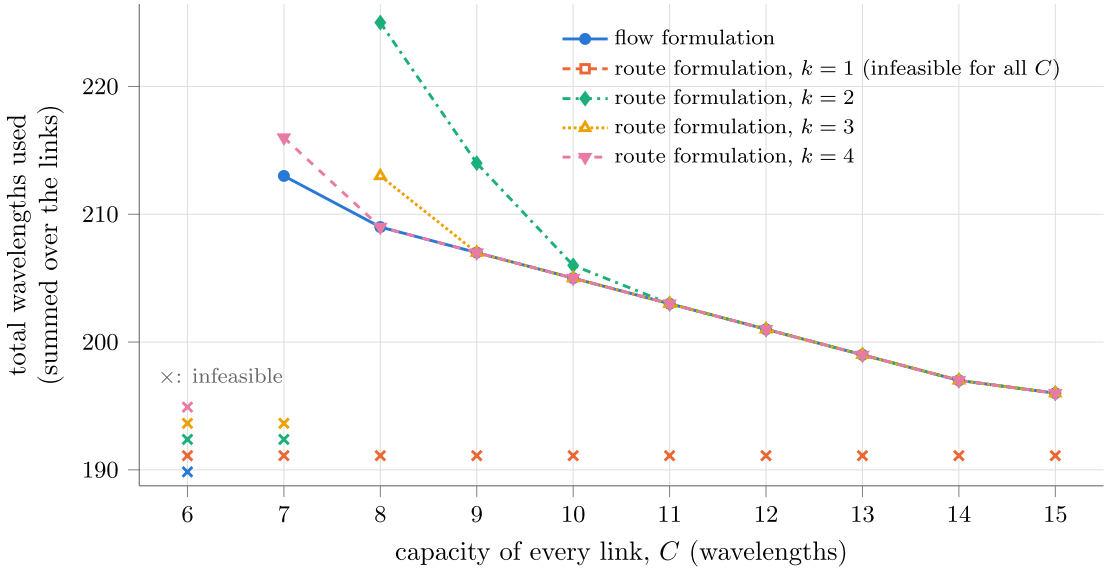


Figure 1. COST239: optimal objective of the flow formulation and of the route formulation for k = 1 to 4. The crosses at the bottom mark the capacities without a solution.

The fewer the wavelengths available, the more candidate paths are needed. For C ≥ 11 two paths per demand give the optimum of the flow formulation. For C = 8 the route formulation needs 225 wavelengths with k = 2, 213 with k = 3 and 209, the optimum of the flow formulation, with k = 4. For C = 7, the smallest capacity for which the flow formulation is feasible, it has no solution with k ≤ 3, and with k = 4 it needs 216 wavelengths instead of 213. With k = 1 it is infeasible for every capacity of the figure: the fewest-hop routing needs C = 22.

5. Model size and computation time

The fourth part of the script compares the two models on GEANT2 (35 nodes, 110 links, 60 demands) close to its smallest feasible capacity. As in handout 4, every case is solved 5 times and the times are given as mean ± standard deviation.

```
GEANT2 (real topology #22): 35 nodes, 110 links, 60 demands, 267 wavelengths
time in seconds: mean ± standard deviation of 5 runs
   C     case objective variables        time s
  18     flow       898      6600   0.49 ± 0.03
  18    k = 3   infeas.       178   0.03 ± 0.00
  18    k = 5   infeas.       294   0.04 ± 0.00
  18    k = 8       898       468   0.08 ± 0.01
```

```
  20     flow       884      6600   0.49 ± 0.02
  20    k = 3   infeas.       178   0.03 ± 0.00
  20    k = 5       888       294   0.05 ± 0.02
  20    k = 8       884       468   0.08 ± 0.01
  22     flow       878      6600   0.48 ± 0.02
  22    k = 3       878       178   0.03 ± 0.00
  22    k = 5       878       294   0.04 ± 0.00
  22    k = 8       878       468   0.09 ± 0.02
  24     flow       876      6600   0.50 ± 0.04
  24    k = 3       876       178   0.04 ± 0.01
  24    k = 5       876       294   0.05 ± 0.01
  24    k = 8       876       468   0.08 ± 0.01
figure: results/lab2_route_formulation/lab2r_time_large.png
figure: results/lab2_route_formulation/lab2r_time_large.pdf

figures saved in results/lab2_route_formulation
```

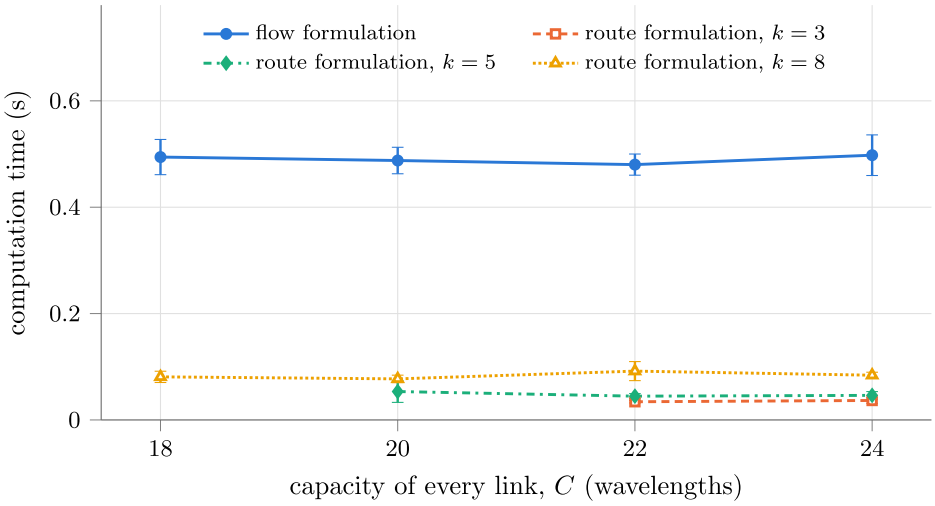


Figure 2. Computation times on GEANT2: mean of 5 runs, with error bars of one standard deviation; only the cases with a solution are shown. The values depend on the computer and on the solver.

The flow formulation has 60 × 110 = 6600 variables, the route formulation with k = 8 has 468, and it is solved about six times faster; the difference is much larger than the standard deviations. Near the smallest feasible capacity, however, the route formulation needs many candidate paths: for C = 18 it is infeasible with k = 5 and reaches the optimum 898 of the flow formulation only with k = 8. In practice the route formulation is preferred on large networks, with a number of candidate paths chosen by checking, as done here, that a larger k does not improve the result.

6. Questions

For every result, record the network, the traffic file, the capacity, the value of k and the solver used.

1. The route formulation has no flow conservation constraints. What guarantees that the wavelengths of a demand travel from its origin to its destination?
2. Find the smallest capacity for which the route formulation with k = 1 is feasible on Spain-7 with the traffic of this lab, using *Compare (sweep)…* with `k=1` and capacities `50-60`. Explain the value with handout 4, section 4.
3. For the same data, can the route formulation give a smaller objective than the flow formulation? Can it give a larger one? Answer by comparing the sets of feasible solutions of the two models.

4. Write the number of variables and of constraints of the two formulations as functions of |N|, |E|, |D| and k, and compute them for GEANT2 with k = 8. Why does the script report 468 variables and not $60 \times 8 = 480$?
5. In TODO 1, compute the candidate paths by length in kilometres (`nx.shortest_simple_paths(G, d.origin, d.destination, weight="length")`) and repeat part 3 for C = 8 and C = 10. Does the optimum change? Why can it change, given that the objective counts hops?
6. How would you write a route formulation in which demands cannot be split? Which variables and constraints change? Optionally, write it in a new file of `formulations/`, starting from `formulations/_template.py`, and compare it with the unsplittable flow formulation of handout 4.